AMATH/PMATH 331: Applied Real Analysis Fall 2026

Assignment 3a; due Tuesday, October 20, 2026

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[1] Let $f\colon A \subseteq \mathbb{R}^n \to \mathbb{R}^m$.

[a] Let $B \subseteq \mathbb{R}^m$. Show that $f^{-1}(B^c) = A \cap (f^{-1}(B))^c$.

[b] Prove that f is continuous on A if and only if:

$$\text{for every } \textit{closed} \text{ subset } F \subseteq \mathbb{R}^m, \text{ then } f^{-1}(F) = A \cap E \text{ for some } \textit{closed} \text{ subset } E \subseteq \mathbb{R}^n. \quad (1)$$

(Note that if the domain A is all of $\mathbb{R}^n$, we can simplify this to $f^{-1}(F)$ is closed in $\mathbb{R}^n$ for every F closed in $\mathbb{R}^m$.)

[c] Show that $B = \{(x, y) \in \mathbb{R}^2 : xy = 1\}$ is a closed subset of $\mathbb{R}^2$. *Hint:* Consider $g(x, y) = xy$.

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[2] Let $f\colon A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be a map. For $\mathbf{x} \in A$, we have

$$f(\mathbf{x}) = \big(f_1(\mathbf{x}), \ldots, f_m(\mathbf{x})\big),$$

where $f_j\colon A \subseteq \mathbb{R}^n \to \mathbb{R}$ is the jth *component function* of f, for $1 \leq j \leq m$.

[a] Let $\mathbf{x}_0 \in A$. Prove using the sequential characterization of continuity that f is continuous at $\mathbf{x}_0$ if and only if each f_j is continuous at $\mathbf{x_0}$ for all $1 \leq j \leq m$.

[b] Let $\pi_j\colon \mathbb{R}^m \to \mathbb{R}$ be the map $\pi_j(y_1, \ldots, y_m) = y_j$, the projection onto the jth component. We proved in class directly using the definition of continuity that π_j is continuous on $\mathbb{R}^m$. Let $\iota_j\colon \mathbb{R} \to \mathbb{R}^m$ be the map $\iota_j(t) = (0, \ldots 0, t, 0, \ldots, 0)$ where the t is in the jth position. Prove directly using the definition of continuity that ι_j is continuous on $\mathbb{R}$.

[c] Find a way to use the maps π_j, ι_j in [b] to give another proof that f is continuous on A if and only if each f_j is continuous on A for $1 \leq j \leq m$.

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[3] Let $\mathbb{R}^{n^2}$ denote the space of $n \times n$ (square) matrices. Recall we can multiply two $n \times n$ matrices A, B to obtain a matrix AB.

[a] Let $A, B \in \mathbb{R}^{n^2}$. Recall that the Euclidean inner product on this space is $\langle A, B\rangle = \sum_{i=1}^n \sum_{j=1}^n A_{ij}B_{ij}$. Prove that $\langle A, B\rangle = \operatorname{tr}(A^T B)$, where A^T is the transpose of A, and $\operatorname{tr}(C)$ is the trace of C.

[b] Consider the map $g\colon \mathbb{R}^{n^2} \to \mathbb{R}^{n^2}$ given by $g(A) = A^T A$. Prove that g is continuous on $\mathbb{R}^{n^2}$.

[c] Prove that the subset of $n \times n$ *orthogonal matrices* is a *compact* subset of $\mathbb{R}^{n^2}$.

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[4] Let $A \subseteq \mathbb{R}^n$ be nonempty, and fix a point $\mathbf{x} \in \mathbb{R}^n$.

[a] If A is *compact*, show that there exists at least one point $\mathbf{y} \in A$ such that $\|\mathbf{y} - \mathbf{x}\| \geq \|\mathbf{a} - \mathbf{x}\|$ for all $\mathbf{a} \in A$. That is, $\mathbf{y}$ is a point in A that is *furthest* from $\mathbf{x}$.

Hint: Find a way to use the Extreme Value Theorem.

[b] Give an example to show that $\mathbf{y}$ is not unique in general. (There are many possible answers.)

[c] If A is *not compact*, there may not exist any such $\mathbf{y}$. If A is noncompact because it is unbounded, then this is trivial to see, as there are points in A arbitrarily far from the origin, and hence arbitrarily far from any $\mathbf{x}$. More interesting is if A is noncompact because it is bounded but *not closed*. Give an example of such an A and $\mathbf{x}$ such that there is no farthest point in A from $\mathbf{x}$. (There are many possible answers.)

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[5] Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be continuous on $\mathbb{R}^n$. Suppose that f has the following property: for any $\varepsilon > 0$, there exists $M > 0$ such that if $\|\mathbf{x}\| > M$, then $|f(\mathbf{x})| < \varepsilon$. (We say that f *goes to zero at infinity.*)

[a] Show that f is bounded on $\mathbb{R}^n$.

[b] Suppose there exists some point $\mathbf{x}_0 \in \mathbb{R}^n$ such that $f(\mathbf{x}_0) > 0$. Prove that there exists a point $\tilde{\mathbf{x}}$ such that $f(\mathbf{x}) \leq f(\tilde{\mathbf{x}})$ for all $\mathbf{x} \in \mathbb{R}^n$. (That is, show that f *attains a global maximum* on $\mathbb{R}^n$.)
[Note that we cannot *directly* use the Extreme Value Theorem because $\mathbb{R}^n$ is not compact. Find a creative way to use it nevertheless.]

[c] Show that the assumption that there exists $\mathbf{x}_0 \in \mathbb{R}^n$ with $f(\mathbf{x}_0) > 0$ is necessary for part [b]. That is, find an example where f *goes to zero at infinity*, with $f(\mathbf{x}) \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n$, and such that f does not attain a global maximum on $\mathbb{R}^n$. [Take $n = 1$ for simplicity.]
